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Problematic of high-performance applications



What are your main challenges and concerns when selecting microcontroller for your high-performance application?

We asked our customers..

Main decision makers for microcontroller selection

Real-time performance

Memory constraints & scalability

Product integration

Ecosystem for rapid development

Product innovation

Product temperature range

MCU

Addressing security

Large package offer

Longevity & Supply chain

Development support

Solution cost

Family consistency

How to address challenges?





Extensive STM32H7 portfolio



Dual-core Line

STM32H745/755
480+240 MHz **SMPS**
1027 + 300 DMIPS
RAM 1 MB
Flash up to 2 MB

STM32H747/757
480+240 MHz **SMPS**
1027 + 300 DMIPS
RAM 1 MB
Flash up to 2 MB



Single-core Line

STM32H7A3/B3
280 MHz **LDO**
599 DMIPS
RAM 1.4 MB
Flash up to 2 MB

STM32H742
480 MHz **LDO**
1027 DMIPS
RAM 692 KB
Flash up to 2 MB

STM32H743/753
480 MHz **LDO**
1027 DMIPS
RAM 1 MB
Flash up to 2 MB

STM32H723/733
550 MHz **LDO**
1177 DMIPS
RAM 564 KB
Flash up to 1 MB

STM32H725/735
550 MHz **SMPS**
1177 DMIPS
RAM 564 KB
Flash up to 1 MB



Extended temperature
range 125 °C ambient



Value Line

STM32H7B0
280 MHz **LDO**
599 DMIPS
RAM 1.4 MB
Flash 128 KB

STM32H750
480 MHz **LDO**
1027 DMIPS
RAM 1 MB
Flash 128 KB

STM32H730
550 MHz **LDO**
1177 DMIPS
RAM 564 KB
Flash 128 KB

STM32H730Q
550 MHz **SMPS**
1177 DMIPS
RAM 564 KB
Flash 128 KB

Arm® Cortex® core

Cortex-M7

Cortex-M7 & -M4

Real time performance

Customer is looking for real-time performance and deterministic application behaviour

Absolute performance 2778 coremarks
Highest performant single-core device with embedded flash
Test code available and results published on EEMBC website

Deterministic behavior: code execution from internal memories
featuring 64kB of L1 cache for data and instructions

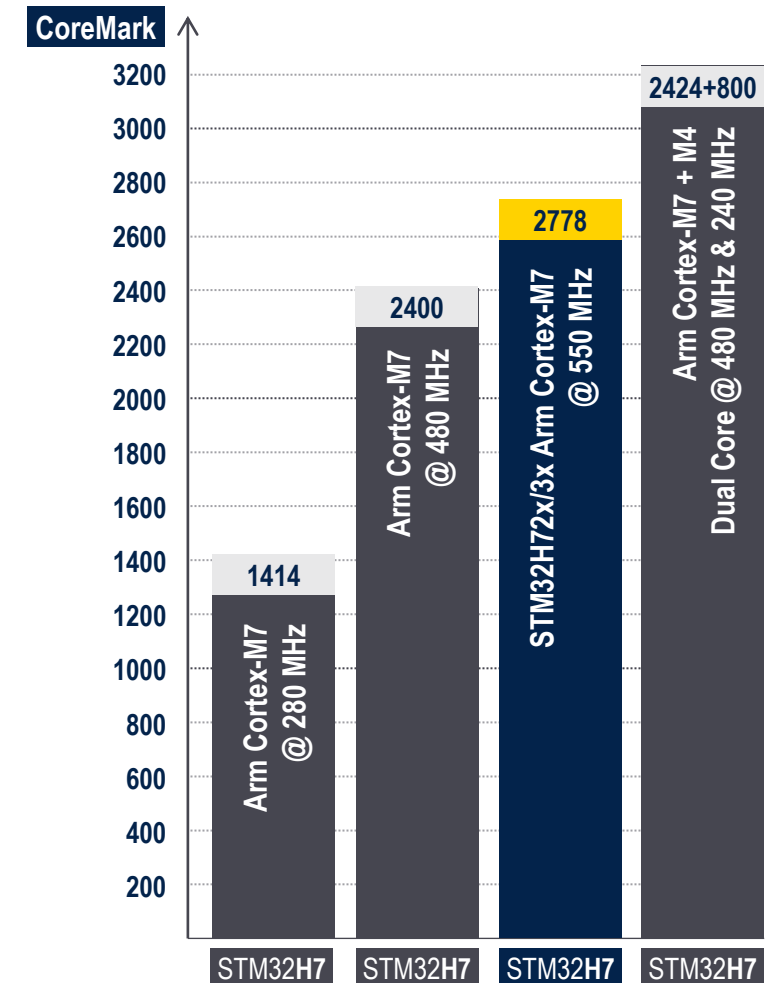
Multiple bus matrixes for autonomous operation
Each dedicated for maximum performance, scalability and no-compromise, each with its own DMA controllers

D1 domain with ARM Cortex-M7
to deliver maximum performance

Includes high speed peripherals

D2 autonomous domain for
connectivity with its own RAM
memory to offload CPU and its bus
matrix

Low power D3 domain to allow data
acquisition in lower modes



Rich integrated peripherals

STM32H72x/3x block diagram

Rich set of integration peripherals without compromise

Autonomous peripherals with DMA access to offload CPU

3 bus matrixes to provide maximum CPU performance

3x FDCAN with time-triggered function

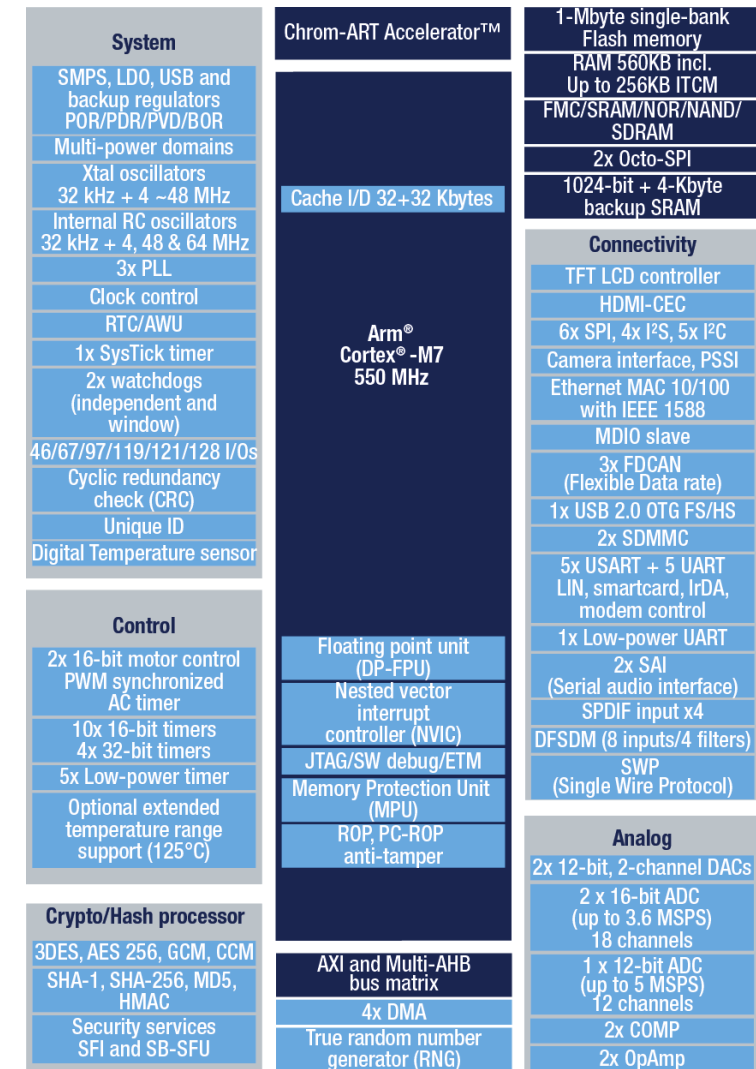
Camera and LCD hardware interfaces with dedicated DMA
Starting with smallest packages

Ethernet controller

External memories with optional on-the-fly decryption

3x up to 16-bit ADC with 19-channels each

Application oriented features, motor control timers, state-of-the-art security, USB



Package offer

STM32H72x/3x

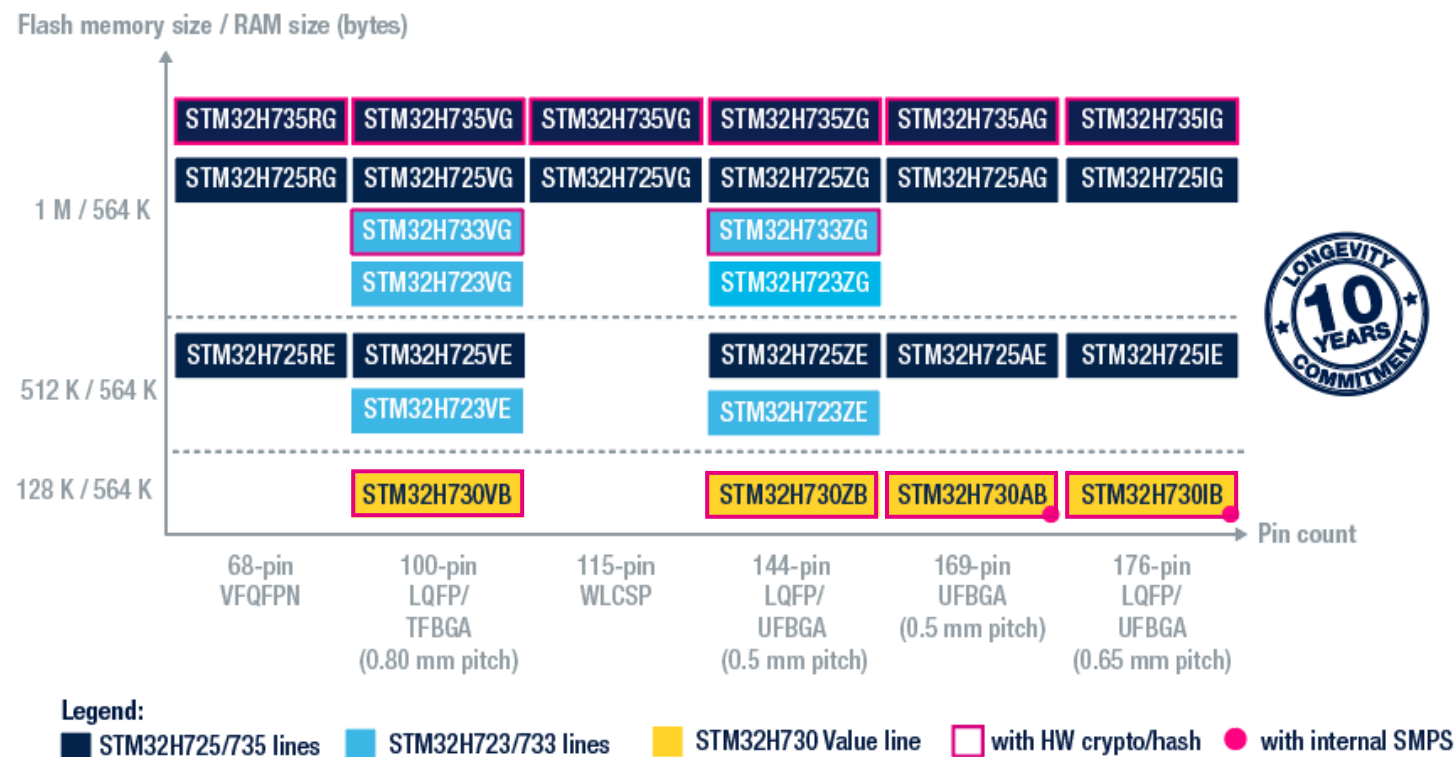
Extensive package offer starting at 68 pins for affordable low-cost 4-layer PCB

Down to VFQFPN68 package and up to LQFP/BGA 176

Industrial ready BGA packages

Pin to pin compatibility between different flash variants

Select package to fit your application, no compromise on performance



Cost of the solution

Customer is looking for most cost optimized solution

Optimized packages to allow 4-layer designs

Wide input voltage range 1.62-3.6V w/o PMU and w/ no additional external active components

Optimized current consumption and product efficiency with integrated DC-DC converter w/ external passive components only

State-of-the-art internal peripherals, including separate ULP battery domain and precise RTC

STM32H7 is a one-stop shop solution to address cost reduction of ever-growing and demanded applications

Memory architecture of your choice

with STM32H72x/3x

Customer decides architecture for code execution

Internal memory only

1 MB internal flash
564kB internal RAM

External memory only

128 kB internal flash
564kB internal RAM
2x OCTOSPI interface
On the fly decryption

External and internal memory in parallel

1 MB internal flash
564kB internal RAM
2x OCTOSPI interface
On the fly decryption
Simple migration
between architectures

External memory only with internal RAM

128 kB internal flash
564kB internal RAM
for critical routines
2x OCTOSPI interface
On the fly decryption

Application architecture of your choice

Supported external memories

with STM32H72x/3x

Select memory optimized for your application

PSRAM/HyperRAM

Low-cost RAM memory
8-bit data width
High-speed
Easy to route
2x OCTOSPI interface
Small MCU package

SRAM/SDRAM Nand FLASH

Flexible memory
controller
Different parallel memory
configurations
Up to 2x 256MB

NOR flash

Code execution
2x OCTOSPI interface
QUADSPI capable
On-the-fly decryption
Flexible architecture

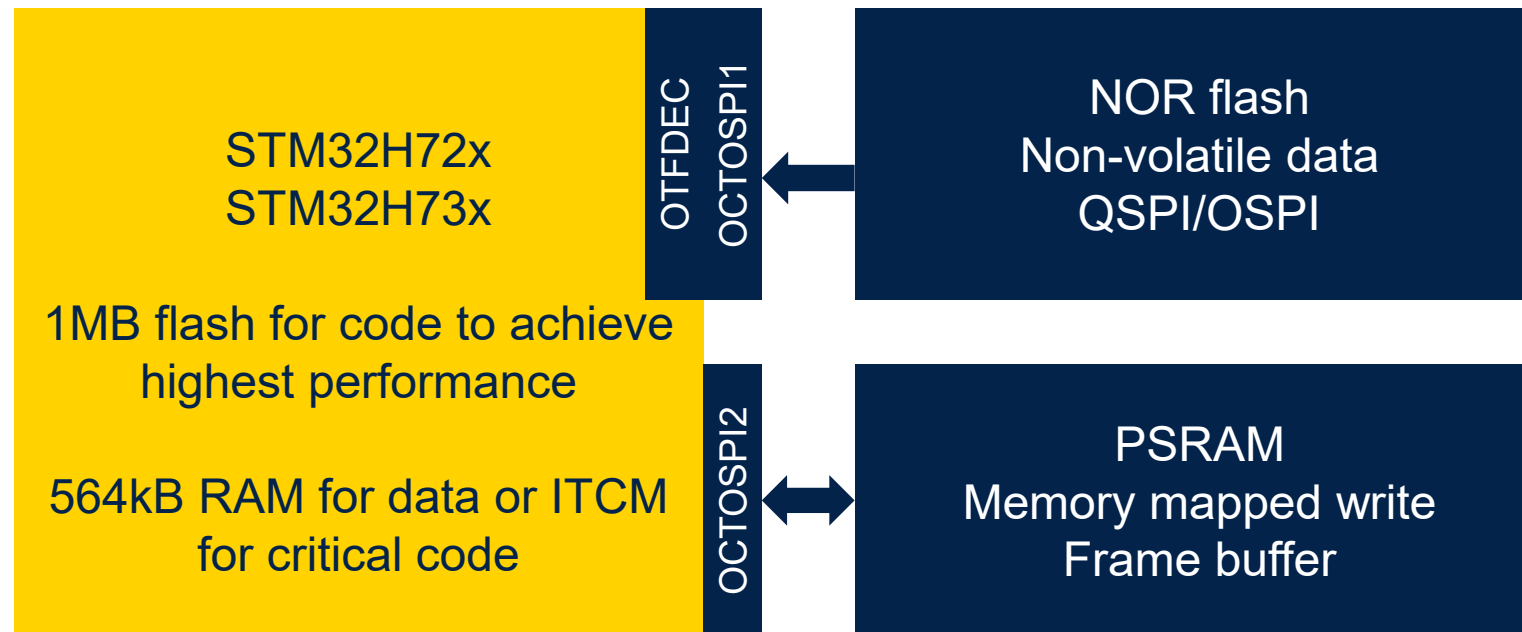
eMMC

2x SDMMC interface
eMMC
SD-card interface

Low-cost external memories setup

with STM32H72x/3x

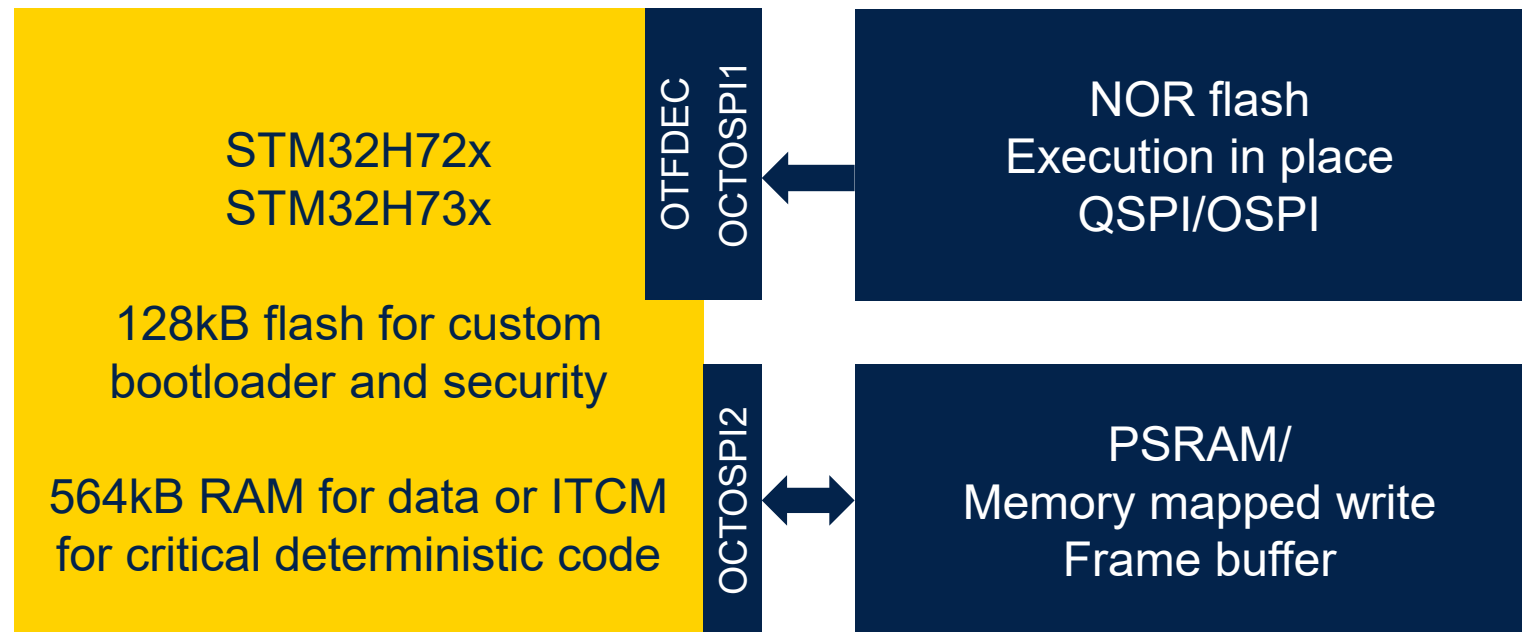
Code executed from internal and external flash with more RAM memory



Low-cost external memories setup

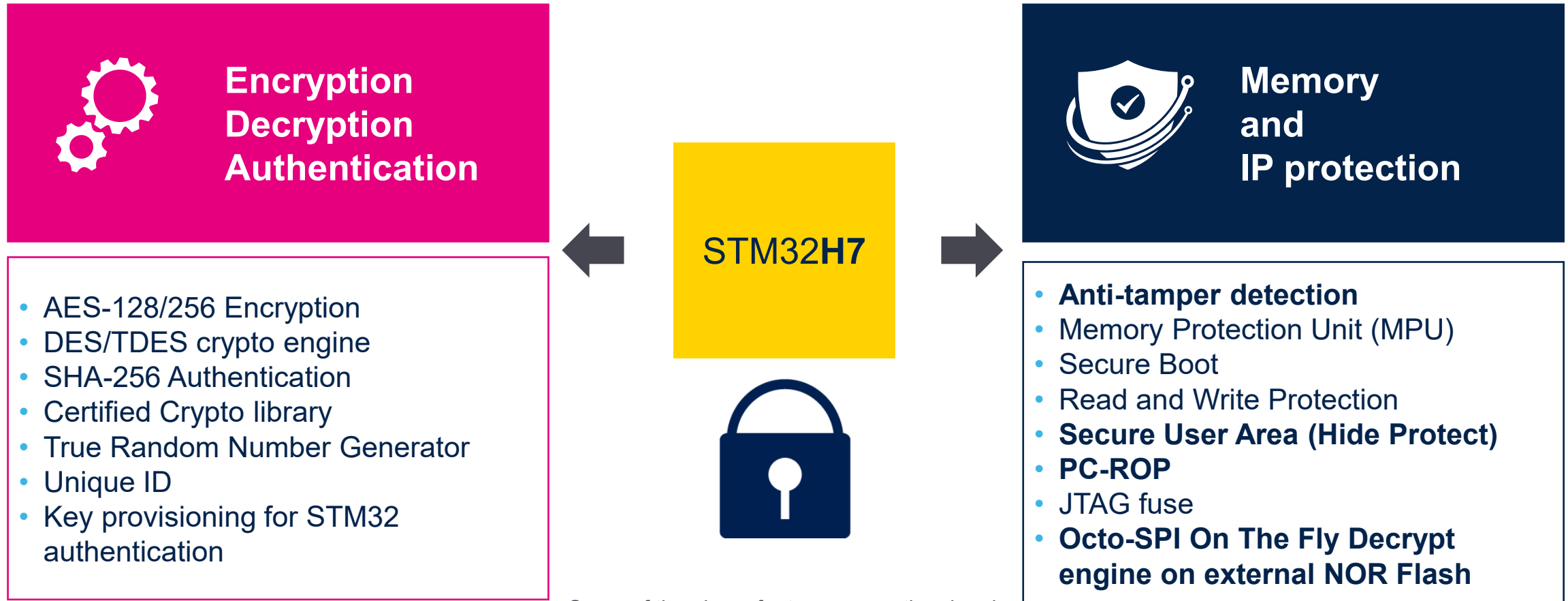
with STM32H72x/3x

Code storage/execution in external flash



Addressing security

No compromise on security offer



Some of the above features are optional and require to procure dedicated part numbers.
Please refer to product specification

Product longevity and supply chain

Product longevity to address industrial and consumer applications

Guaranteed 10 years renewable product longevity

Products today available at least up to 2030

Dual manufacturing sourcing to guarantee QoS

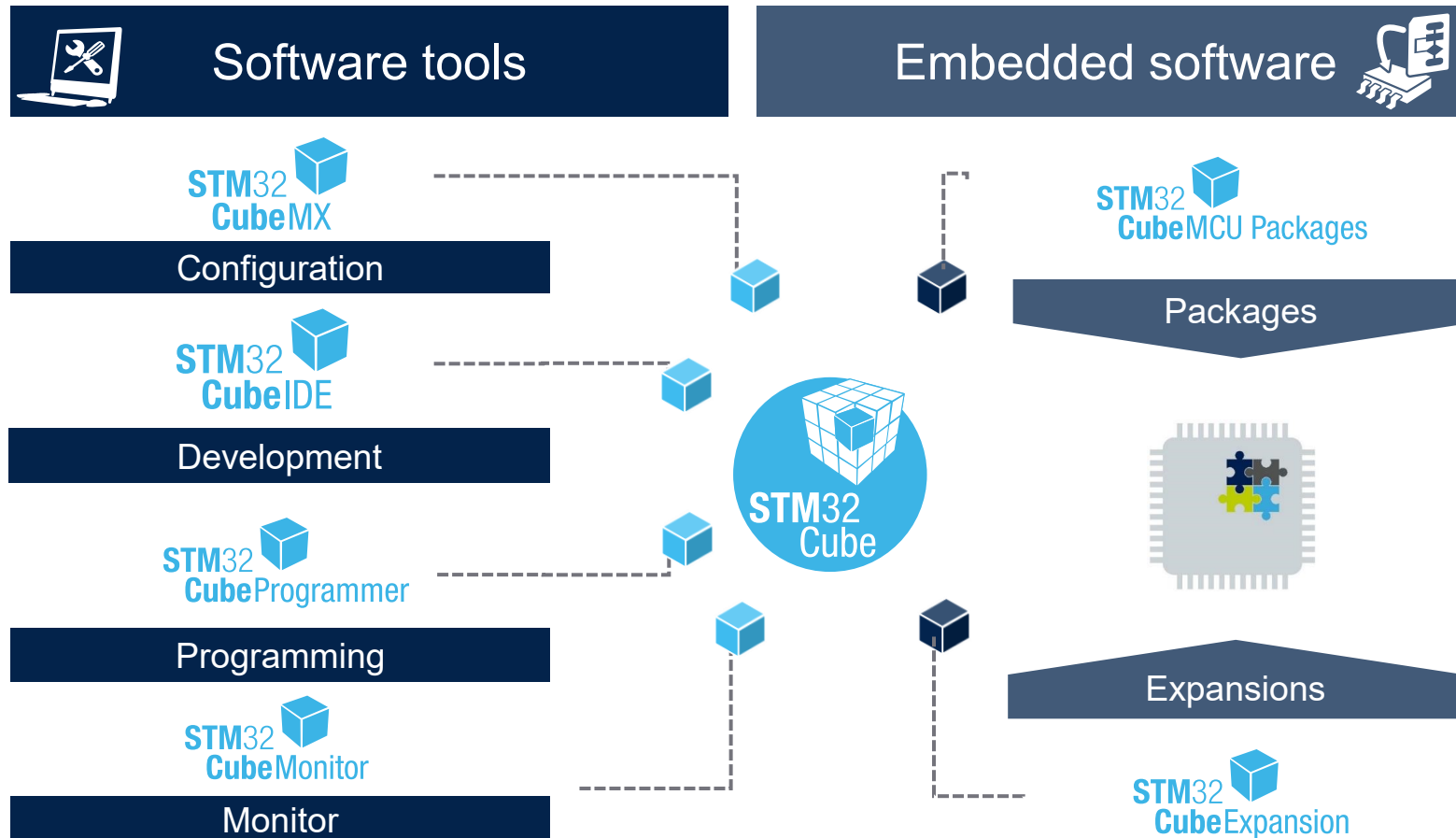
Products available at ST or through distributors and e-stores for immediate second day delivery

- **STM32F1** (launched in **2007**)
- **STM32L1** (launched in **2009**)
- **STM32F2** (launched in **2010**)
- **STM32F4** (launched in **2011**)
- **STM32F0** (launched in **2012**)
- **STM32F3** (launched in **2012**)
- **STM32L0** (launched in **2013**)
- **STM32F7** (launched in **2014**)
- **STM32L4** (launched in **2015**)
- **STM32L4+** (launched in **2016**)
- **STM32H7** (launched in **2016**)
- **STM32WB** (launched in **2018**)
- **STM32G0** (launched in **2018**)
- **STM32MP1** (launched in **2019**)
- **STM32G4** (launched in **2019**)
- **STM32WL** (launched in **2020**)

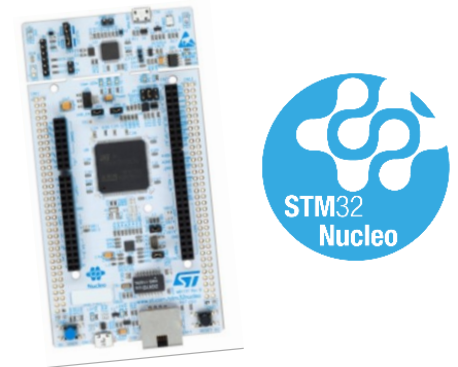
Available between 10 and 23 years

Microcontroller ecosystem

Firmware, software and hardware collaterals for successful application development



STM32 Nucleo-144 boards



Discovery kits



Worldwide support and community

You are never alone with STM32

Worldwide FAE support offered directly by ST

40 application engineers in EMEA dedicated to customers and oriented to application specifics

Online community forum dedicated for customers

Massive open online courses hosted on Youtube

Online trainings or customer specific

More than 300 partners in ST portfolio



FAE - Worldwide
Customer Support



community.st.com

**MO
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Conclusion

Conclusion

STM32 series are the one stop-shop for your next high performance application

Delivers the highest performant microcontroller on a market with embedded flash

Rich memory and IP integration starting with very small packages

STM32 is not limiting our customers with memory architecture

Easy to use with STM32Cube ecosystem

Thank you

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